

WALLET RISK SCORING

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Data Collection Method

The transaction history for each wallet was retrieved using the **Etherscan API**, the industry-standard block explorer for Ethereum. Live on-chain data was fetched by connecting directly to the Ethereum network via an **Alchemy** node.

- **Rationale:** Initial attempts to use The Graph Protocol were met with deprecated and unreliable API endpoints. The Etherscan API offers a more robust method for fetching a wallet's complete transaction history, while a direct node connection is the ground truth for live data.
- **Process:**
 1. A Python script called Etherscan's `account_txlist` endpoint for each wallet address to fetch its entire transaction history.
 2. This raw data was then filtered locally to isolate only the transactions that interacted with official **Compound V2 smart contracts**.
 3. A separate script using the `web3.py` library connected to an Alchemy node to call the `getAccountLiquidity` function on the Compound V2 Comptroller contract, retrieving the live health factor for each wallet.

This two-pronged approach ensures both historical completeness and real-time accuracy and is highly scalable.

Feature Selection Rationale

The features were strategically chosen to create a comprehensive risk profile by blending a user's **historical behavior** with their **live, on-chain financial state**. This ensures the model is both backward-looking (evaluating past performance) and forward-looking (assessing current risk).

- **health_factor_usd (Live Data):** This is the most critical feature. It represents the wallet's real-time financial health on the protocol, fetched directly from the Compound smart contracts. It is a direct measure of the account's collateral value relative to its borrow value. A low Health Factor (approaching zero from the positive side) signifies an imminent risk of liquidation, making it the most powerful predictive indicator.
- **liquidation_count (Historical Data):** This is the most important historical indicator. A past liquidation is a definitive, unambiguous sign of poor risk management and an inability to handle market volatility. It is a direct measure of past failure.

- **account_age_days (Historical Data):** This measures the time since the wallet's first interaction with Compound. A longer history can suggest a more experienced user and provides more data to assess their long-term behavior.
- **transaction_count (Historical Data):** This serves as a proxy for user activity. While not a direct risk signal on its own, it provides context. Very low activity could indicate a "set-and-forget" position, which is inherently risky. High activity suggests a user who is actively managing their position.

Normalization Method

To compare these diverse features, each was normalized to a common scale of **0 to 1** using the **Min-Max Scaling** technique. This standard statistical method ensures that each feature contributes proportionally to the final score without its original scale (e.g., days vs. dollars) skewing the result.

The formula for Min-Max Scaling is:

$$X_{\text{norm}} = \frac{X_{\text{max}} - X_{\text{min}}}{X - X_{\text{min}}}$$

A crucial step in the process is to orient the normalized scores so that **a value of 1 always represents maximum risk**.

- For features where a high value is inherently risky (e.g., liquidation_count), the normalized score is used directly.
- For features where a high value is beneficial (e.g., health_factor_usd, account_age_days), the score is **inverted** after normalization: $\text{Risk} = 1 - X_{\text{norm}}$.

Scoring Logic

A **weighted scoring model** was used to aggregate the normalized features into a final risk score from **0 (Lowest Risk)** to **1000 (Highest Risk)**. This approach allows us to assign more importance to the most predictive features.

1. **Weighting:** Each normalized feature was assigned a weight based on its importance in determining immediate risk. The live Health Factor is given the most significance.
 - **Health Factor:** 50%
 - **Liquidation Count:** 30%
 - **Account Age:** 10%
 - **Transaction Count:** 10%
2. **Aggregation:** The final score was calculated as the weighted average of the normalized risk values for each feature.

3. **Scaling:** This final weighted average (a value between 0 and 1) was multiplied by 1000 to produce the final score in the desired 0-1000 range.

The complete formula is:

Final Score = (w_health * norm_health + w_liq * norm_liq + w_age * norm_age + w_tx * norm_tx) * 1000

Justification of Risk Indicators

The chosen indicators provide a comprehensive and defensible measure of risk for several reasons:

- **Forward-Looking and Backward-Looking:** The model is not one-dimensional. It balances **historical behavior** (liquidation_count, account_age) with a **forward-looking, real-time assessment** (health_factor). This prevents a user with a previously poor record but a currently safe position from being unfairly penalized, and vice-versa.
- **Protocol-Native Metrics:** By using the **Health Factor**, we are directly leveraging Compound's own internal risk calculation, which is the ground truth for liquidation events on the protocol.
- **Clear Behavioral Signals:** A liquidation event is an unambiguous signal of risk. It's not a proxy but a direct observation of a user's position failing under market stress.